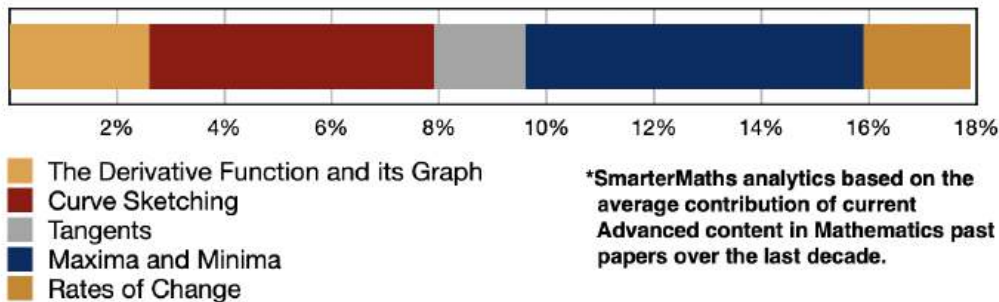


ADV: Calculus (Adv), C3 Applications of Calculus (Adv) Maxima and Minima (Y12)

Teacher: Ljiljana Todric

Exam Equivalent Time: 172.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C3 Applications of Calculus



HISTORICAL CONTRIBUTION

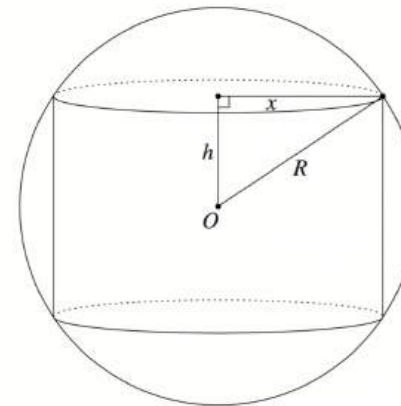
- *C3 Applications of Calculus* is the biggest topic in the Advanced course, contributing a massive 17.9% to past Mathematics exams, on average over the past decade.
- This topic has been split into five sub-topics for analysis purposes: 1-*The Derivative Function and its Graph* (2.6%), 2-*Curve Sketching* (5.3%), 3-*Tangents* (1.7%), 4-*Maxima and Minima* (6.3%) and 5-*Rates of Change* (2.0%).
- This analysis looks at the largest sub-topic *Maxima and Minima*

HSC ANALYSIS - What to expect and common pitfalls

- *Maxima and Minima* (6.3%) is the biggest sub-topic within the largest topic area. It appears in every exam with significant mark allocations. In short, its importance in achieving a band 5-6 result cannot be overstated.
- *Max/Min* problems will typically ask students to prove an equation in the earlier parts. This forms the basis of later questions and typically is at a Band 4 level - a good opportunity for all students to score well.
- Important exam strategy: HSC Markers consistently remind students to work with the given equation of the question, *even if they couldn't successfully complete the proof of the equation in the earlier part*.
- *Max/Min* problems have underlying themes. *Area/Volume equations* have dominated (8 times in the last decade, including each year 2018-2020). Other themes include *distance* (2017 and 2013) and a *cost equation* (2009).
- *Max/Min* problems that involve topics such as plane geometry which are no longer in the syllabus, have been removed from the database.

Questions

1. Calculus, 2ADV C3 2005 HSC 8a



A cylinder of radius x and height $2h$ is to be inscribed in a sphere of radius R centred at O as shown.

i. Show that the volume of the cylinder is given by

$$V = 2\pi h(R^2 - h^2). \quad (1 \text{ mark})$$

ii. Hence, or otherwise, show that the cylinder has a maximum volume when $h = \frac{R}{\sqrt{3}}$. (3 marks)

2. Calculus, 2ADV C3 2014 HSC 16c

The diagram shows a window consisting of two sections. The top section is a semicircle of diameter x m. The bottom section is a rectangle of width x m and height y m.

The entire frame of the window, including the piece that separates the two sections, is made using 10 m of thin metal.



The semicircular section is made of coloured glass and the rectangular section is made of clear glass.

Under test conditions the amount of light coming through one square metre of the coloured glass is 1 unit and the amount of light coming through one square metre of the clear glass is 3 units.

The total amount of light coming through the window under test conditions is L units.

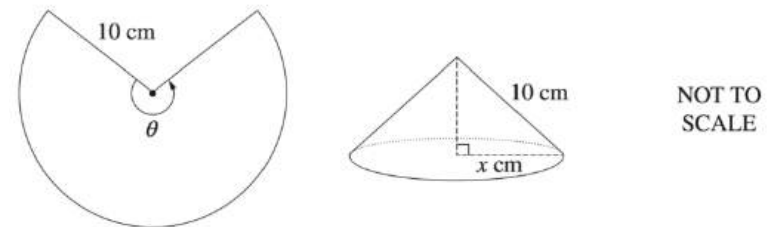
i. Show that $y = 5 - x\left(1 + \frac{\pi}{4}\right)$. (2 marks)

ii. Show that $L = 15x - x^2\left(3 + \frac{5\pi}{8}\right)$. (2 marks)

iii. Find the values of x and y that maximise the amount of light coming through the window under test conditions. (3 marks)

3. Calculus, 2ADV C3 2018 HSC 16a

A sector with radius 10 cm and angle θ is used to form the curved surface of a cone with base radius x cm, as shown in the diagram.



The volume of a cone of radius r and height h is given by $V = \frac{1}{3}\pi r^2 h$.

i. Show that the volume, V cm³, of the cone described above is given by

$$V = \frac{1}{3}\pi x^2 \sqrt{100 - x^2}. \quad (1 \text{ mark})$$

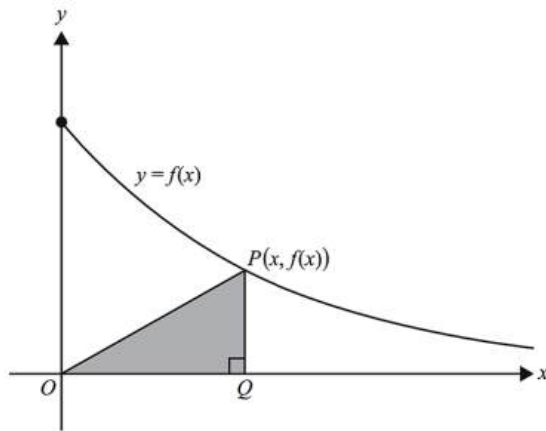
ii. Show that $\frac{dV}{dx} = \frac{\pi x(200 - 3x^2)}{3\sqrt{100 - x^2}}$. (2 marks)

iii. Find the exact value of θ for which V is a maximum. (3 marks)

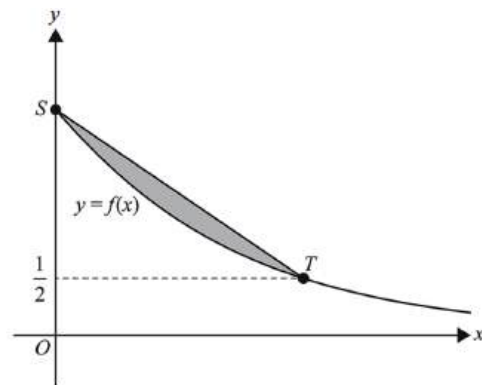
4. Calculus, 2ADV C4 SM-Bank 12

Let $f(x) = 2e^{-\frac{x}{5}}$ for $x \geq 0$

A right-angled triangle OQP has vertex O at the origin, vertex Q on the x -axis and vertex P on the graph of f , as shown. The coordinates of P are $(x, f(x))$.



- Find the area, A , of the triangle OPQ in terms of x . (1 mark)
- Find the maximum area of triangle OQP and the value of x for which the maximum occurs. (3 marks)
- Let S be the point on the graph of f on the y -axis and let T be the point on the graph of f with the y -coordinate $\frac{1}{2}$. Find the area of the region bounded by the graph of f and the line segment ST . (2 marks)

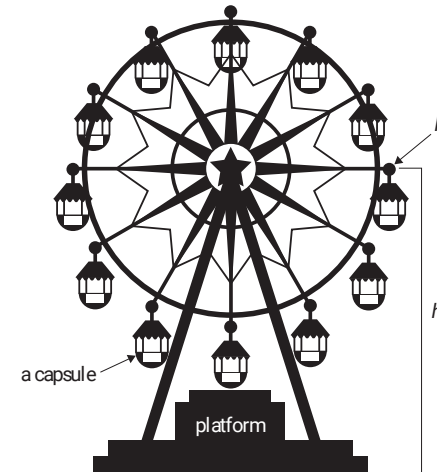


5. Trigonometry, 2ADV T3 SM-Bank 16

Sammy visits a giant Ferris wheel. Sammy enters a capsule on the Ferris wheel from a platform above the ground. The Ferris wheel is rotating anticlockwise. The capsule is attached to the Ferris wheel at point P . The height of P above the ground, h , is modelled by

$h(t) = 65 - 55 \cos\left(\frac{\pi t}{15}\right)$, where t is the time in minutes after Sammy enters the capsule and h is measured in metres.

Sammy exits the capsule after one complete rotation of the Ferris wheel.



- State the minimum and maximum heights of P above the ground. (1 mark)
- For how much time is Sammy in the capsule? (1 mark)
- Find the rate of change of h with respect to t and, hence, state the value of t at which the rate of change of h is at its maximum. (2 marks)

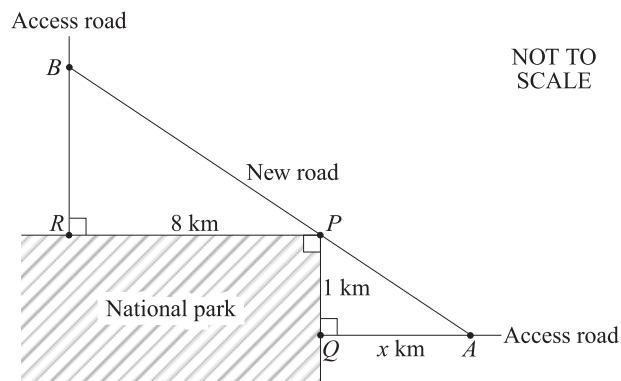
6. Calculus, 2ADV C3 2019 HSC 15c

The entry points, R and Q , to a national park can be reached via two straight access roads. The access roads meet the national park boundaries at right angles. The corner, P , of the national park is 8 km from R and 1 km from Q . The boundaries of the national park form a right angle at P .

A new straight road is to be built joining these roads and passing through P .

Points A and B on the access roads are to be chosen to minimise the distance, D km, from A to B along the new road.

Let the distance QA be x km.



i. Show that $D^2 = (x + 8)^2 + \left(\frac{8}{x} + 1\right)^2$. (3 marks)

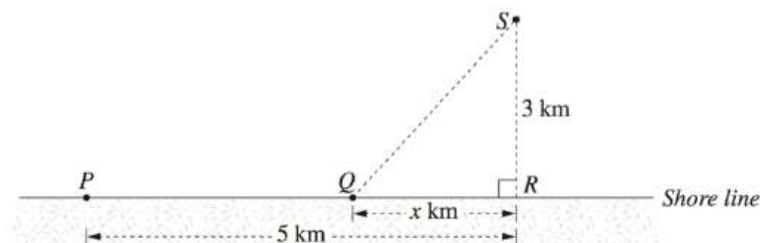
ii. Show that $x = 2$ gives the minimum value of D^2 . (3 marks)

7. Calculus, 2ADV C3 2009 HSC 9b

An oil rig, S , is 3 km offshore. A power station, P , is on the shore. A cable is to be laid from P to S . It costs \$1000 per kilometre to lay the cable along the shore and \$2600 per kilometre to lay the cable underwater from the shore to S .

The point R is the point on the shore closest to S , and the distance PR is 5 km.

The point Q is on the shore, at a distance of x km from R , as shown in the diagram.

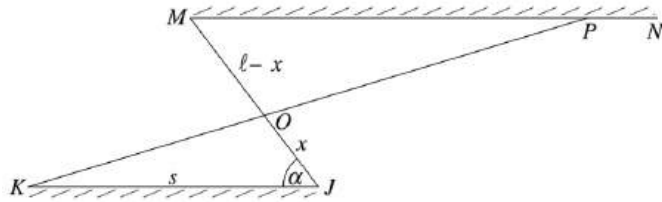


- Find the total cost of laying the cable in a straight line from P to R and then in a straight line from R to S . (1 mark)
- Find the cost of laying the cable in a straight line from P to S . (1 mark)
- Let $\$C$ be the total cost of laying the cable in a straight line from P to Q , and then in a straight line from Q to S .

Show that $C = 1000(5 - x + 2.6\sqrt{x^2 + 9})$. (2 marks)

- Find the minimum cost of laying the cable. (4 marks)
- New technology means that the cost of laying the cable underwater can be reduced to \$1100 per kilometre.
Determine the path for laying the cable in order to minimise the cost in this case. (2 marks)

8. Calculus, 2ADV C3 2008 HSC 10b



The diagram shows two parallel brick walls KJ and MN joined by a fence from J to M . The wall KJ is s metres long and $\angle KJM = \alpha$. The fence JM is l metres long.

A new fence is to be built from K to a point P somewhere on MN . The new fence KP will cross the original fence JM at O .

Let $OJ = x$ metres, where $0 < x < l$.

i. Show that the total area, A square metres, enclosed by $\triangle OKJ$ and $\triangle OMP$ is given by

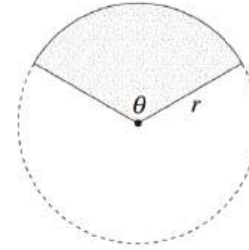
$$A = s \left(x - l + \frac{l^2}{2x} \right) \sin \alpha. \quad (3 \text{ marks})$$

ii. Find the value of x that makes A as small as possible. Justify the fact that this value of x gives the minimum value for A . (3 marks)

iii. Hence, find the length of MP when A is as small as possible. (1 mark)

9. Calculus, 2ADV C3 2011 HSC 10b

A farmer is fencing a paddock using P metres of fencing. The paddock is to be in the shape of a sector of a circle with radius r and sector angle θ in radians, as shown in the diagram.



i. Show that the length of fencing required to fence the perimeter of the paddock is

$$P = r(\theta + 2). \quad (1 \text{ mark})$$

ii. Show that the area of the sector is $A = \frac{1}{2}Pr - r^2$. (1 mark)

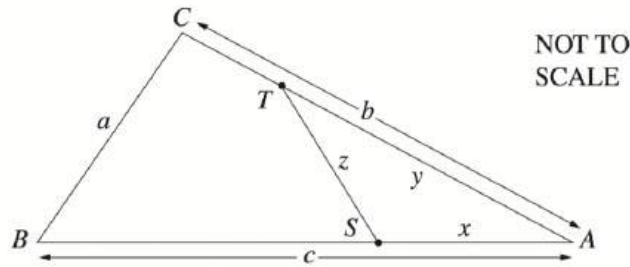
iii. Find the radius of the sector, in terms of P , that will maximise the area of the paddock. (2 marks)

iv. Find the angle θ that gives the maximum area of the paddock. (1 mark)

v. Explain why it is only possible to construct a paddock in the shape of a sector if

$$\frac{P}{2(\pi + 1)} < r < \frac{P}{2} \quad (2 \text{ marks})$$

10. Calculus, 2ADV C3 2004 HSC 10b



The diagram shows a triangular piece of land ABC with dimensions $AB = c$ metres, $AC = b$ metres and $BC = a$ metres, where $a \leq b \leq c$.

The owner of the land wants to build a straight fence to divide the land into two pieces of equal area. Let S and T be points on AB and AC respectively so that ST divides the land into two pieces of equal area.

Let $AS = x$ metres, $AT = y$ metres and $ST = z$ metres.

i. Show that $xy = \frac{1}{2}bc$. (1 mark)

ii. Use the cosine rule in triangle AST to show that

$$z^2 = x^2 + \frac{b^2 c^2}{4x^2} - bc \cos A. \quad (2 \text{ marks})$$

iii. Show that the value of z^2 in the equation in part (ii) is a minimum when

$$x = \sqrt{\frac{bc}{2}}. \quad (4 \text{ marks})$$

iv. Show that the minimum length of the fence is $\sqrt{\frac{(P-2b)(P-2c)}{2}}$ metres, where

$$P = a + b + c.$$

(You may assume that the value of x given in part (iii) is feasible.) (2 marks)

11. Calculus, 2ADV C3 2007 HSC 10b

The noise level, N , at a distance d metres from a single sound source of loudness L is given by the formula

$$N = \frac{L}{d^2}.$$

Two sound sources, of loudness L_1 and L_2 are placed m metres apart.



The point P lies on the line between the sound sources and is x metres from the sound source with loudness L_1 .

i. Write down a formula for the sum of the noise levels at P in terms of x . (1 mark)

ii. There is a point on the line between the sound sources where the sum of the noise levels is a minimum.

Find an expression for x in terms of m , L_1 and L_2 if P is chosen to be this point. (4 marks)

12. Calculus, 2ADV C3 2017 HSC 16a

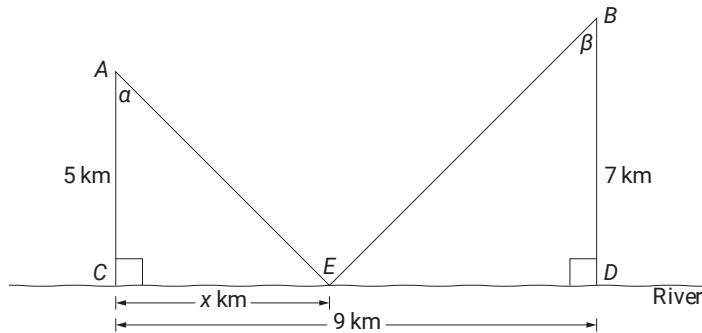
John's home is at point **A** and his school is at point **B**. A straight river runs nearby.

The point on the river closest to **A** is point **C**, which is 5 km from **A**.

The point on the river closest to **B** is point **D**, which is 7 km from **B**.

The distance from **C** to **D** is 9 km.

To get some exercise, John cycles from home directly to point **E** on the river, **x** km from **C**, before cycling directly to school at **B**, as shown in the diagram.



The total distance John cycles from home to school is **L** km.

i. Show that $L = \sqrt{x^2 + 25} + \sqrt{49 + (9 - x)^2}$. (1 mark)

ii. Show that if $\frac{dL}{dx} = 0$, then $\sin \alpha = \sin \beta$. (3 marks)

iii. Find the value of **x** that makes $\sin \alpha = \sin \beta$. (2 marks)

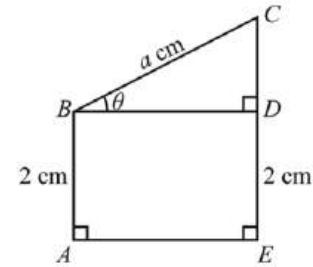
iv. Explain why this value of **x** gives a minimum for **L**. (1 mark)

13. Calculus, 2ADV C3 SM-Bank 13

The figure shown represents a wire frame where **ABCE** is a convex quadrilateral. The point **D** is on line segment **EC** with **AB = ED = 2 cm** and **BC = a cm**, where **a** is a positive constant.

$$\angle BAE = \angle CEA = \frac{\pi}{2}$$

Let $\angle CBD = \theta$ where $0 < \theta < \frac{\pi}{2}$.



i. Find **BD** and **CD** in terms of **a** and θ . (2 marks)

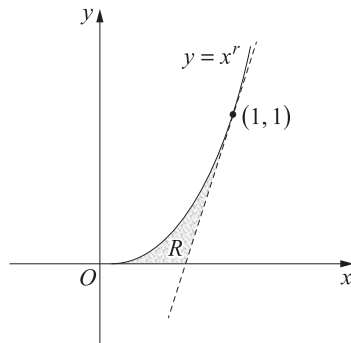
ii. Find the length, **L** cm, of the wire in the frame, including length **BD**, in terms of **a** and θ . (1 mark)

iii. Find $\frac{dL}{d\theta}$, and hence show that $\frac{dL}{d\theta} = 0$ when $BD = 2CD$. (2 marks)

iv. Find the maximum value of **L** if $a = 3\sqrt{5}$. (1 mark)

14. Calculus, 2ADV C4 2019 HSC 16c

The diagram shows the region R , bounded by the curve $y = x^r$, where $r \geq 1$, the x -axis and the tangent to the curve at the point $(1, 1)$.



- i. Show that the tangent to the curve at $(1, 1)$ meets the x -axis at

$$\left(\frac{r-1}{r}, 0\right). \quad (2 \text{ marks})$$

- ii. Using the result of part (i), or otherwise, show that the area of the region R is

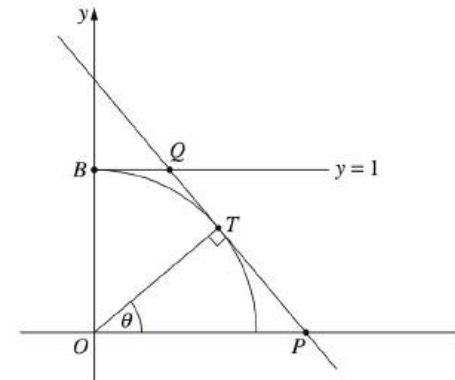
$$\frac{r-1}{2r(r+1)}. \quad (2 \text{ marks})$$

- iii. Find the exact value of r for which the area of R is a maximum. (3 marks)
-

15. Calculus, 2ADV C3 2012 HSC 16b

The diagram shows a point T on the unit circle $x^2 + y^2 = 1$ at an angle θ from the positive x -axis, where $0 < \theta < \frac{\pi}{2}$.

The tangent to the circle at T is perpendicular to OT , and intersects the x -axis at P , and the line $y = 1$ intersects the y -axis at B .



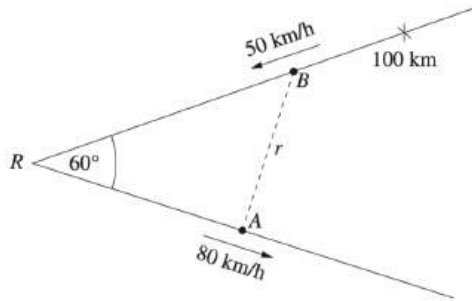
- i. Show that the equation of the line PT is $x \cos \theta + y \sin \theta = 1$. (2 marks)
 ii. Find the length of BQ in terms of θ . (1 mark)
 iii. Show that the area, A , of the trapezium $OPQB$ is given by

$$A = \frac{2 - \sin \theta}{2 \cos \theta} \quad (2 \text{ marks})$$

- iv. Find the angle θ that gives the minimum area of the trapezium. (3 marks)
-

16. Calculus, 2ADV C3 2013 HSC 14b

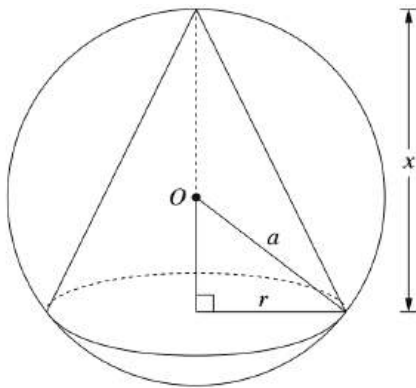
Two straight roads meet at R at an angle of 60° . At time $t = 0$ car A leaves R on one road, and car B is 100 km from R on the other road. Car A travels away from R at a speed of 80 km/h, and car B travels towards R at a speed of 50 km/h.



The distance between the cars at time t hours is r km.

- Show that $r^2 = 12\,900t^2 - 18\,000t + 10\,000$. (2 marks)
- Find the minimum distance between the cars. (3 marks)

17. Calculus, 2ADV C3 2006 HSC 9c



A cone is inscribed in a sphere of radius a , centred at O . The height of the cone is x and the radius of the base is r , as shown in the diagram.

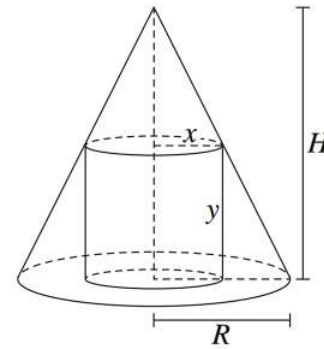
- Show that the volume, V , of the cone is given by

$$V = \frac{1}{3}\pi(2ax^2 - x^3). \quad (2 \text{ marks})$$

- Find the value of x for which the volume of the cone is a maximum. You must give reasons why your value of x gives the maximum volume. (3 marks)

18. Calculus, 2ADV C3 2015 HSC 16c

The diagram shows a cylinder of radius x and height y inscribed in a cone of radius R and height H , where R and H are constants.



The volume of a cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.

The volume of a cylinder of radius r and height h is $\pi r^2 h$.

- Show that the volume, V , of the cylinder can be written as

$$V = \frac{H}{R}\pi x^2(R - x). \quad (3 \text{ marks})$$

- By considering the inscribed cylinder of maximum volume, show that the volume of any inscribed cylinder does not exceed $\frac{4}{9}$ of the volume of the cone. (4 marks)

Worked Solutions

1. Calculus, 2ADV C3 2005 HSC 8a

$$\begin{aligned}\text{i. } V &= \pi r^2 h \quad (\text{general form}) \\ &= \pi x^2 \times 2h\end{aligned}$$

Using Pythagoras

$$\begin{aligned}R^2 &= h^2 + x^2 \\ x^2 &= R^2 - h^2 \\ \therefore V &= 2\pi h(R^2 - h^2) \quad \dots \text{as required.}\end{aligned}$$

$$\begin{aligned}\text{ii. } V &= 2\pi h(R^2 - h^2) \\ &= 2\pi R^2 h - 2\pi h^3 \\ \frac{dV}{dh} &= 2\pi R^2 - 3 \times 2\pi h^2 \\ &= 2\pi R^2 - 6\pi h^2\end{aligned}$$

$$\frac{d^2V}{dh^2} = -12\pi h$$

$$\text{Max or min when } \frac{dV}{dh} = 0$$

$$2\pi R^2 - 6\pi h^2 = 0$$

$$6\pi h^2 = 2\pi R^2$$

$$h^2 = \frac{R^2}{3}$$

$$h = \frac{R}{\sqrt{3}}, \quad h > 0$$

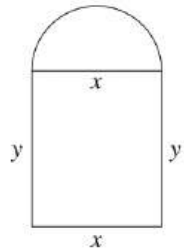
$$\text{When } h = \frac{R}{\sqrt{3}}$$

$$\frac{d^2V}{dh^2} = -12\pi \times \frac{R}{\sqrt{3}} < 0$$

$$\therefore \text{Volume is a maximum when } h = \frac{R}{\sqrt{3}} \text{ units.}$$

2. Calculus, 2ADV C3 2014 HSC 16c

i.



Frame is 10m

$$10 = 2x + 2y + \frac{1}{2}\pi x$$

$$2y = 10 - 2x - \frac{1}{2}\pi x$$

$$\begin{aligned}\therefore y &= 5 - x - \frac{\pi}{4}x \\ &= 5 - x\left(1 + \frac{\pi}{4}\right) \quad \dots \text{as required.}\end{aligned}$$

ii. Area (clear) = $x \times y$

$$\begin{aligned}\text{Area (colour)} &= \frac{1}{2} \times \pi r^2 \\ &= \frac{1}{2} \times \pi \left(\frac{x}{2}\right)^2 \\ &= \frac{\pi x^2}{8}\end{aligned}$$

$$\begin{aligned}\therefore L &= 3xy + \left(\frac{\pi x^2}{8} \times 1\right) \\ &= 3x\left[5 - x\left(1 + \frac{\pi}{4}\right)\right] + \frac{\pi x^2}{8} \quad (\text{see part (i)}) \\ &= 15x - 3x^2 - \frac{3x^2\pi}{4} + \frac{x^2\pi}{8} \\ &= 15x - 3x^2 - \frac{5x^2\pi}{8} \\ &= 15x - x^2\left(3 + \frac{5\pi}{8}\right) \quad \dots \text{as required}\end{aligned}$$

♦ Mean mark 35%

iii.
$$L = 15x - x^2\left(3 + \frac{5\pi}{8}\right)$$

$$\frac{dL}{dx} = 15 - 2x\left(3 + \frac{5\pi}{8}\right)$$

$$\frac{d^2L}{dx^2} = -2\left(3 + \frac{5\pi}{8}\right)$$

Max or min when $\frac{dL}{dx} = 0$

$$15 - 2x\left(3 + \frac{5\pi}{8}\right) = 0$$

$$2x\left(3 + \frac{5\pi}{8}\right) = 15$$

$$\begin{aligned}x &= \frac{15}{2\left(3 + \frac{5\pi}{8}\right)} \\ &= 1.51103\dots \\ &= 1.511 \quad (3 \text{ d.p.})\end{aligned}$$

Since $\frac{d^2L}{dx^2} < 0 \Rightarrow \text{MAX}$

When $x = 1.511$

$$\begin{aligned}y &= 5 - 1.511\left(1 + \frac{\pi}{4}\right) \\ &= 2.3022\dots \\ &= 2.302 \quad (3 \text{ d.p.})\end{aligned}$$

$\therefore$ MAX light when $x = 1.511$ m
and $y = 2.302$ m.

♦ Mean mark 38%

COMMENT: A sanity check for your answer could be to compare your answers to the perimeter restriction of 10m.

3. Calculus, 2ADV C3 2018 HSC 16a

i. $V = \frac{1}{3}\pi r^2 h$

Using Pythagoras,

$$h = \sqrt{100 - x^2}$$

$$r = x$$

$$\therefore \text{Volume} = \frac{1}{3}\pi x^2 \sqrt{100 - x^2}$$

♦ Mean mark 45%.

ii. $V = \frac{1}{3}\pi x^2 \sqrt{100 - x^2}$

$$\frac{dV}{dh} = \frac{1}{3}\pi \left[2x \cdot \sqrt{100 - x^2} - 2x \cdot \frac{1}{2}(100 - x^2)^{-\frac{1}{2}} \cdot x^2 \right]$$

$$= \frac{1}{3}\pi \left[\frac{2x(100 - x^2) - x^3}{\sqrt{100 - x^2}} \right]$$

$$= \frac{1}{3}\pi \left[\frac{200x - 2x^3 - x^3}{\sqrt{100 - x^2}} \right]$$

$$= \frac{\pi x(200 - 3x^2)}{3\sqrt{100 - x^2}} \text{ .. as required}$$

iii. Find x when $\frac{dV}{dx} = 0$

$$200 - 3x^2 = 0$$

$$x = \sqrt{\frac{200}{3}}$$

When $x < \sqrt{\frac{200}{3}}$, $(200 - 3x^2) > 0$

$$\Rightarrow \frac{dV}{dx} > 0$$

When $x > \sqrt{\frac{200}{3}}$, $(200 - 3x^2) < 0$

$$\Rightarrow \frac{dV}{dx} < 0$$

♦♦ Mean mark 23%.

$$\therefore \text{MAX when } x = \sqrt{\frac{200}{3}}$$

Equating the arc length of the section
to the circumference of the cone:

$$2\pi r \cdot \frac{\theta}{2\pi} = 2\pi \cdot x$$

$$10\theta = 2\pi \sqrt{\frac{200}{3}}$$

$$\begin{aligned} \therefore \theta &= \frac{2\pi \cdot 10\sqrt{2}}{10 \cdot \sqrt{3}} \\ &= \frac{2\sqrt{2}\pi}{\sqrt{3}} \end{aligned}$$

4. Calculus, 2ADV C4 SM-Bank 12

$$\begin{aligned} \text{i. Area} &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} x \left(2e^{-\frac{x}{5}} \right) \\ \therefore A &= xe^{-\frac{x}{5}} \end{aligned}$$

ii. Stationary point when $\frac{dA}{dx} = 0$,

$$\begin{aligned} x \left(-\frac{1}{5} e^{-\frac{x}{5}} \right) + e^{-\frac{x}{5}} &= 0 \\ e^{-\frac{x}{5}} \left(1 - \frac{x}{5} \right) &= 0 \\ \therefore x &= 5 \quad \left(e^{-\frac{x}{5}} > 0, \text{ for all } x \right) \end{aligned}$$

$$\text{When } x = 5, A = xe^{-\frac{x}{5}} = 5e^{-1}$$

$$\therefore A_{\max} = \frac{5}{e} \text{ u}^2, \text{ when } x = 5$$

iii. Find S : $F(0) = 2$

$$\Rightarrow S(0, 2)$$

$$\text{Find } T: 2e^{-\frac{x}{5}} = \frac{1}{2}$$

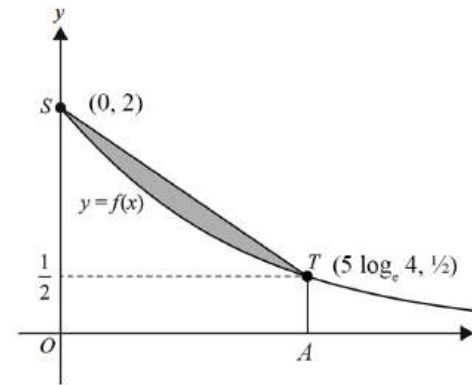
$$e^{-\frac{x}{5}} = \frac{1}{4}$$

$$-\frac{x}{5} = \log_e \left(\frac{1}{4} \right)$$

$$x = 5 \log_e(4)$$

$$\Rightarrow T \left(5 \log_e(4), \frac{1}{2} \right)$$

♦ Mean mark (Vic) 35%.



$$\begin{aligned} \therefore \text{Area} &= \text{Area } SOAT - \int_0^{5 \log_e(4)} \left(2e^{-\frac{x}{5}} \right) dx \\ &= \frac{1}{2} h(a+b) + 10 \left[e^{-\frac{x}{5}} \right]_0^{5 \log_e(4)} \\ &= \frac{5}{2} \log_e(4) \left(2 + \frac{1}{2} \right) + 10 \left[e^{-\log_e(4)} - e^0 \right] \\ &= \frac{25}{4} \log_e(4) + 10 \left(\frac{1}{4} - 1 \right) \\ &= \frac{25}{4} \log_e(4) - \frac{15}{2} \text{ u}^2 \end{aligned}$$

♦♦ Mean mark (Vic) 32%.

5. Trigonometry, 2ADV T3 SM-Bank 16

$$\begin{aligned} \text{i. } h_{\min} &= 65 - 55 & h_{\max} &= 65 + 55 \\ &= 10 \text{ m} & &= 120 \text{ m} \end{aligned}$$

$$\text{ii. Period} = \frac{2\pi}{\frac{\pi}{15}} = 30 \text{ min}$$

$$\text{iii. } h'(t) = 65 - 55 \cos\left(\frac{\pi t}{15}\right)$$

$$\begin{aligned} h'(t) &= \frac{\pi}{15} \times 55 \sin\left(\frac{\pi}{15}t\right) \\ &= \frac{11\pi}{3} \sin\left(\frac{\pi}{15}t\right) \end{aligned}$$

$$\text{Since } \sin\left(\frac{\pi}{15}t\right)_{\max} = \sin\left(\frac{\pi}{2}\right),$$

$\therefore h'(t)_{\max}$ occurs when

$$\frac{\pi t}{15} = \frac{\pi}{2}$$

$$\begin{aligned} \therefore t &= \frac{\pi}{2} \times \frac{15}{\pi} \\ &= \frac{15}{2} \text{ minutes } (0 \leq t \leq 30) \end{aligned}$$

6. Calculus, 2ADV C3 2019 HSC 15c

$$\begin{aligned} \text{i. } \angle BRP &= \angle PQA = 90^\circ \text{ (given)} \\ \angle BPR &= \angle PAQ \text{ (corresponding)} \\ \Rightarrow \triangle BRP &\parallel \triangle PQA \text{ (equiangular)} \end{aligned}$$

$$\frac{BR}{RP} = \frac{PQ}{QA} \quad (\text{corresponding sides of similar triangles})$$

$$\frac{BR}{8} = \frac{1}{x}$$

$$BR = \frac{8}{x}$$

Using Pythagoras,

$$\begin{aligned} D^2 &= (QA + RP)^2 + (BR + PQ)^2 \\ &= (x + 8)^2 + \left(\frac{8}{x} + 1\right)^2 \end{aligned}$$

♦ Mean mark part (i) 41 %.

$$\text{ii. } (D^2)' = 2 \cdot (x + 8) + 2(-1) \cdot \frac{8}{x^2} \left(\frac{8}{x} + 1\right)$$

$$= 2x + 16 - \frac{16}{x^2} \left(\frac{8}{x} + 1\right)$$

$$= 2x + 16 - \frac{128}{x^3} - \frac{16}{x^2}$$

$$(D^2)'' = 2 - (-3) \cdot \frac{128}{x^4} - (-2) \cdot \frac{16}{x^3}$$

$$= 2 + \frac{384}{x^4} + \frac{32}{x^3}$$

When $x = 2$,

$$(D^2)' = 2 \times 2 + 16 - \frac{128}{8} - \frac{16}{4} = 0$$

$\Rightarrow$ S.P. at $x = 2$

$$(D^2)'' = 2 + \frac{384}{16} + \frac{32}{8} > 0$$

$\therefore$ MIN value of D^2 at $x = 2$.

♦♦ Mean mark part (ii) 31 %.

7. Calculus, 2ADV C3 2009 HSC 9b

$$\begin{aligned}\text{i. Cost} &= (PR \times 1000) + (SR \times 2600) \\ &= (5 \times 1000) + (3 \times 2600) \\ &= 12\,800\end{aligned}$$

$\therefore$ Cost is \$12 800

$$\text{ii. Cost} = PS \times 2600$$

Using Pythagoras:

$$\begin{aligned}PS^2 &= PR^2 + SR^2 \\ &= 5^2 + 3^2 \\ &= 34 \\ PS &= \sqrt{34}\end{aligned}$$

$$\begin{aligned}\therefore \text{Cost} &= \sqrt{34} \times 2600 \\ &= 15\,160.474\dots \\ &= \$15\,160 \text{ (nearest dollar)}\end{aligned}$$

$$\text{iii. Show } C = 1000(5 - x + 2.6\sqrt{x^2 + 9})$$

$$\text{Cost} = (PQ \times 1000) + (QS \times 2600)$$

$$PQ = 5 - x$$

$$\begin{aligned}QS^2 &= QR^2 + SR^2 \\ &= x^2 + 3^2\end{aligned}$$

$$QS = \sqrt{x^2 + 9}$$

$$\therefore C = (5 - x)1000 + \sqrt{x^2 + 9} (2600)$$

$$= 1000(5 - x + 2.6\sqrt{x^2 + 9}) \dots \text{as required}$$

$$\text{iv. Find the MIN cost of laying the cable}$$

$$C = 1000(5 - x + 2.6\sqrt{x^2 + 9})$$

$$\frac{dC}{dx} = 1000\left(-1 + 2.6 \times \frac{1}{2} \times 2x(x^2 + 9)^{-\frac{1}{2}}\right)$$

♦♦ Although specific data is unavailable for question parts, mean marks were 35% for Q9 in total.

$$= 1000\left(-1 + \frac{2.6x}{\sqrt{x^2 + 9}}\right)$$

$$\text{MAX/MIN when } \frac{dC}{dx} = 0$$

$$1000\left(-1 + \frac{2.6x}{\sqrt{x^2 + 9}}\right) = 0$$

$$\frac{2.6x}{\sqrt{x^2 + 9}} = 1$$

$$2.6x = \sqrt{x^2 + 9}$$

$$(2.6)^2 x^2 = x^2 + 9$$

$$x^2(2.6^2 - 1) = 9$$

$$x^2 = \frac{9}{5.76}$$

$$= 1.5625$$

$$x = 1.25 \quad (x > 0)$$

$$\text{If } x = 1, \frac{dC}{dx} < 0$$

$$\text{If } x = 2, \frac{dC}{dx} > 0$$

$$\therefore \text{MIN when } x = 1.25$$

$$C = 1000(5 - 1.25 + 2.6\sqrt{1.25^2 + 9})$$

$$= 1000(122)$$

$$= 12\,200$$

$$\therefore \text{MIN cost is } \$12\,200 \text{ when } x = 1.25$$

$$\text{v. Underwater cable now costs \$1100 per km}$$

$$\Rightarrow C = 1000(5 - x) + 1100\sqrt{x^2 + 9}$$

$$= 1000(5 - x + 1.1\sqrt{x^2 + 9})$$

$$\frac{dC}{dx} = 1000\left(-1 + 1.1 \times \frac{1}{2} \times 2x(x^2 + 9)^{-\frac{1}{2}}\right)$$

$$= 1000\left(-1 + \frac{1.1x}{\sqrt{x^2 + 9}}\right)$$

IMPORTANT: Tougher derivative questions often require students to deal with multiple algebraic constants. See Worked Solutions in part (iv).

MARKER'S COMMENT: Check the nature of the critical points in these type of questions. If using the first derivative test, make sure some actual values are substituted in.

MAX/MIN when $\frac{dC}{dx} = 0$

$$1000 \left(-1 + \frac{1.1x}{\sqrt{x^2 + 9}} \right) = 0$$

$$\frac{1.1x}{\sqrt{x^2 + 9}} = 1$$

$$1.1x = \sqrt{x^2 + 9}$$

$$1.1^2 x^2 = x^2 + 9$$

$$x^2 (1.1^2 - 1) = 9$$

$$x^2 = \frac{9}{0.21}$$

$$x \approx 6.5 \text{ km (to 1 d.p.)}$$

$\Rightarrow$ no solution since $x \leq 5$

If we lay cable PR then RS

$$\Rightarrow \text{Cost} = 5 \times 1100 + 3 \times 1000 = 8500$$

If we lay cable directly underwater via PS

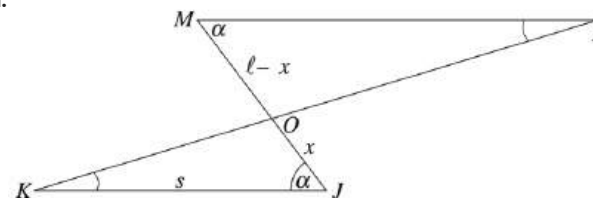
$$\Rightarrow \text{Cost} = \sqrt{34} \times 1100 = 6414.047...$$

$\therefore$ MIN cost is \$6414 by cabling directly from P to S .

MARKER'S COMMENT: Many students failed to interpret a correct calculation of $x > 5$ as providing no solution.

8. Calculus, 2ADV C3 2008 HSC 10b

i.



$$A = \text{Area } \triangle OJK + \text{Area } \triangle OMP$$

Using sine rule

$$\text{Area } \triangle OJK = \frac{1}{2} xs \sin \alpha$$

Area $\triangle OMP \Rightarrow$ Need to find MP

$$\angle OKJ = \angle MPO \text{ (alternate angles, } MP \parallel KJ)$$

$$\angle PMO = \angle OJK = \alpha \text{ (alternate angles, } MP \parallel KJ)$$

$$\therefore \triangle OJK \parallel \triangle OMP \text{ (equiangular)}$$

$$\Rightarrow \frac{x}{s} = \frac{l-x}{MP} \text{ (corresponding sides of similar triangles)}$$

$$MP = \frac{l-x}{x} \cdot s$$

$$\text{Area } \triangle OMP = \frac{1}{2} (l-x) \cdot MP \cdot \sin \alpha$$

$$= \frac{1}{2} (l-x) \cdot \frac{(l-x)}{x} \cdot s \sin \alpha$$

$$\therefore A = \frac{1}{2} x \cdot s \sin \alpha + \frac{1}{2} (l-x) \cdot \frac{(l-x)}{x} \cdot s \sin \alpha$$

$$= s \sin \alpha \left(\frac{1}{2} x + \frac{1}{2} (l-x) \cdot \frac{(l-x)}{x} \right)$$

$$= s \sin \alpha \left(\frac{1}{2} x + \frac{(l-x)^2}{2x} \right)$$

$$= s \sin \alpha \left(\frac{1}{2} x + \frac{l^2}{2x} - l + \frac{1}{2} x \right)$$

$$= s \left(x - l + \frac{l^2}{2x} \right) \sin \alpha \quad \dots \text{ as required}$$

♦♦♦ Low mean marks highlighted (although exact data not available before 2009).

ii. Find x such that A is a minimum

$$A = s \left(x - l + \frac{l^2}{2x} \right) \sin \alpha$$

$$\frac{dA}{dx} = s \left(1 - \frac{l^2}{2x^2} \right) \sin \alpha$$

$$\text{MAX/MIN when } \frac{dA}{dx} = 0$$

$$s \left(1 - \frac{l^2}{2x^2} \right) \sin \alpha = 0$$

$$\frac{l^2}{2x^2} = 1$$

$$2x^2 = l^2$$

$$x^2 = \frac{l^2}{2}$$

$$x = \frac{l}{\sqrt{2}}, \quad x > 0$$

$$\frac{d^2 A}{dx^2} = s \left(\frac{l^2}{2x^3} \right) \sin \alpha$$

Since $0 < \alpha < 90^\circ \Rightarrow \sin \alpha > 0$, $l > 0$ and $x > 0$

$$\frac{d^2 A}{dx^2} > 0 \Rightarrow \text{MIN at } x = \frac{l}{\sqrt{2}}$$

iii. Since $MP = \frac{(l-x)}{x}s$ and MIN when $x = \frac{l}{\sqrt{2}}$

$$MP = \left(\frac{l - \frac{l}{\sqrt{2}}}{\frac{l}{\sqrt{2}}} \right) s \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{(\sqrt{2}l - l)}{l} s$$

$$= (\sqrt{2} - 1)s \text{ metres}$$

$\therefore MP = (\sqrt{2} - 1)s$ metres when A is a MIN.

MARKER'S COMMENT: Students who could not complete part (i) are reminded that they can still proceed to part (ii) and attempt to differentiate the result given. Note that l and α are constants when differentiating.

9. Calculus, 2ADV C3 2011 HSC 10b

i. Need to show $P = r(\theta + 2)$

$$\begin{aligned} P &= (2 \times r) + \frac{\theta}{2\pi} \times 2\pi r \\ &= 2r + r\theta \\ &= r(\theta + 2) \dots \text{as required} \end{aligned}$$

ii. Need to show $A = \frac{1}{2}Pr - r^2$.

$$\text{Since } P = r(\theta + 2) \Rightarrow \theta = \frac{P - 2r}{r}$$

$$\begin{aligned} \therefore A &= \frac{1}{2}r^2\theta \\ &= \frac{1}{2}r^2 \cdot \frac{P - 2r}{r} \\ &= \frac{1}{2}(Pr - 2r^2) \\ &= \frac{1}{2}Pr - r^2 \dots \text{as required} \end{aligned}$$

iii. $A = \frac{1}{2}Pr - r^2$

$$\frac{dA}{dr} = \frac{1}{2}P - 2r$$

MAX or MIN when $\frac{dA}{dr} = 0$

$$\frac{1}{2}P - 2r = 0$$

$$2r = \frac{1}{2}P$$

$$r = \frac{P}{4}$$

$$\frac{d^2A}{dr^2} = -2 \Rightarrow \text{MAX}$$

$\therefore$ Area is a MAX when $r = \frac{P}{4}$ units

♦ Mean mark 41%.

TIP: Area of a sector = $\pi r^2 \times$ the percentage of the circle that the sector angle accounts for, i.e.

$$\frac{\theta}{2\pi} \text{ radians. } \therefore A = \pi r^2 \times \frac{\theta}{2\pi} = \frac{1}{2}r^2\theta$$

♦♦ Mean mark 29%

MARKER'S COMMENT: The second derivative test proved much more successful and easily proven in this part. Make sure you are comfortable choosing between it and the 1st derivative test depending on the required calcs.

iv. Need to find θ when area is a MAX $\Rightarrow r = \frac{P}{4}$

$$\begin{aligned} P &= r(\theta + 2) \\ &= \frac{P}{4}(\theta + 2) \end{aligned}$$

$$4P = P(\theta + 2)$$

$$\theta + 2 = 4$$

$$\theta = 2 \text{ radians}$$

♦♦♦ Mean mark 20%

v. For a sector to exist $0 < \theta < 2\pi$, and $\theta = \frac{P - 2r}{r}$

$$\Rightarrow \frac{P - 2r}{r} > 0$$

$$P - 2r > 0$$

$$r < \frac{P}{2}$$

$$\Rightarrow \frac{P - 2r}{r} < 2\pi$$

$$P - 2r < 2r\pi$$

$$2r\pi + 2r > P$$

$$2r(\pi + 1) > P$$

$$r > \frac{P}{2(\pi + 1)}$$

$$\therefore \frac{P}{2(\pi + 1)} < r < \frac{P}{2} \dots \text{as required}$$

♦♦♦ Mean mark 4%. A BEAST!
MARKER'S COMMENT: When asked to 'explain', students should support their answer with a mathematical argument.

i. Area $\triangle ABC = \frac{1}{2} bc \sin A$

Area $\triangle AST = \frac{1}{2} xy \sin A$

Area $\triangle AST = \frac{1}{2} \times \text{Area } \triangle ABC$ (given)

$$\frac{1}{2} xy \sin A = \frac{1}{2} \times \frac{1}{2} bc \sin A$$

$$xy \sin A = \frac{1}{2} bc \sin A$$

$$\therefore xy = \frac{1}{2} bc \dots \text{as required.}$$

ii. Using the cosine rule in $\triangle AST$,

$$z^2 = x^2 + y^2 - 2xy \cos A$$

Since $xy = \frac{1}{2} bc$

$$\therefore y = \frac{bc}{2x}$$

$$\therefore z^2 = x^2 + \left(\frac{bc}{2x}\right)^2 - 2x\left(\frac{bc}{2x}\right) \cos A$$

$$= x^2 + \frac{b^2 c^2}{4x^2} - bc \cos A$$

iii. $z^2 = x^2 + \left(\frac{b^2 c^2}{4}\right) x^{-2} - bc \cos A$

$$\frac{d(z^2)}{dx} = 2x - \frac{2b^2 c^2}{4} x^{-3} = 2x - \frac{b^2 c^2}{2x^3}$$

$$\frac{d^2(z^2)}{dx^2} = 2 + \frac{3b^2 c^2}{2} x^{-4} = 2 + \frac{3b^2 c^2}{2x^4}$$

SP's occur when $\frac{d(z^2)}{dx} = 0$

$$= 0$$

$$2x - \frac{b^2 c^2}{2x^3}$$

$$4x^4 - b^2 c^2 = 0$$

$$4x^4 = b^2 c^2$$

$$x^4 = \frac{b^2 c^2}{4}$$

$$x^2 = \frac{bc}{2}$$

$$x = \sqrt{\frac{bc}{2}}, \quad (x > 0)$$

$$\frac{d^2(z^2)}{dx^2} = 2 + \frac{3b^2 c^2}{2x^4} > 0 \quad (\text{for all } x)$$

$$\therefore \text{A minimum value of } z^2 \text{ when } x = \sqrt{\frac{bc}{2}}.$$

iv. $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

When $x = \sqrt{\frac{bc}{2}},$

$$\therefore z^2 = x^2 + \frac{b^2 c^2}{4x^2} - bc \cos A$$

$$= \frac{bc}{2} + \frac{b^2 c^2}{4\left(\frac{bc}{2}\right)} - bc \left(\frac{b^2 + c^2 - a^2}{2bc}\right)$$

$$= \frac{bc}{2} + \frac{bc}{2} - \left(\frac{b^2 + c^2 - a^2}{2}\right)$$

$$= \frac{2bc - b^2 - c^2 + a^2}{2}$$

$$= \frac{a^2 - (b^2 - 2bc + c^2)}{2}$$

$$= \frac{1}{2} [a^2 - (b - c)^2]$$

$$= \frac{1}{2} [(a - (b - c))(a + (b - c))]$$

$$= \frac{1}{2} [(a - b + c)(a + b - c)]$$

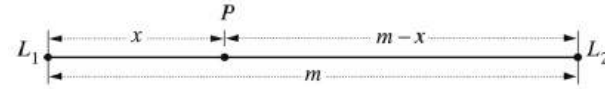
$$= \frac{1}{2}[(a+b+c-2b)(a+b+c-2c)]$$

$$= \frac{(P-2b)(P-2c)}{2} \quad (\text{using } P = a+b+c)$$

$$\therefore z = \sqrt{\frac{(P-2b)(P-2c)}{2}} \text{ metres ... as required.}$$

11. Calculus, 2ADV C3 2007 HSC 10b

i.



$$N = \frac{L}{d^2}$$

$$\text{Noise from } L_1 = \frac{L_1}{x^2}$$

$$\text{Noise from } L_2 = \frac{L_2}{(m-x)^2}$$

$$\therefore N = \frac{L_1}{x^2} + \frac{L_2}{(m-x)^2}$$

ii. $N = L_1 x^{-2} + L_2(m-x)^{-2}$

$$\frac{dN}{dx} = -2L_1x^{-3} + -2L_2(m-x)^{-3} \times \frac{d}{dx}(m-x)$$

$$= \frac{-2L_1}{x^3} + \frac{2L_2}{(m-x)^3}$$

Max or min when $\frac{dN}{dx} = 0$

$$\frac{2L_1}{x^3} = \frac{2L_2}{(m-x)^3}$$

$$2L_1(m-x)^3 = 2L_2x^3$$

$$L_1(m-x)^3 = L_2x^3$$

$$\sqrt[3]{L_1}(m-x) = \sqrt[3]{L_2}x$$

$$\sqrt[3]{L_1}m - \sqrt[3]{L_1}x = \sqrt[3]{L_2}x$$

$$\sqrt[3]{L_2}x + \sqrt[3]{L_1}x = \sqrt[3]{L_1}m$$

$$x(\sqrt[3]{L_2} + \sqrt[3]{L_1}) = \sqrt[3]{L_1}m$$

$$x = \frac{\sqrt[3]{L_1}m}{(\sqrt[3]{L_2} + \sqrt[3]{L_1})}$$

$$= -2L_1x^{-3} + 2L_2(m-x)^{-3}$$

$$\frac{dN}{dx}$$

$$\frac{d^2 N}{dx^2} = 6L_1 x^{-4} - 6L_2(m-x)^{-4} \times -1$$

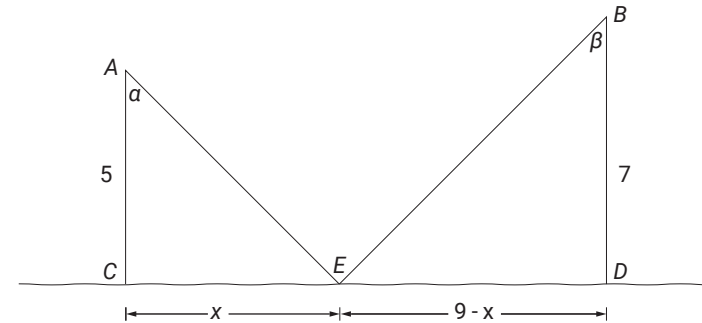
$$= \frac{6L_1}{x^4} + \frac{6L_2}{(m-x)^4} > 0$$

∴ A minimum occurs when

$$x = \frac{\sqrt[3]{L_1} m}{\left(\sqrt[3]{L_2} + \sqrt[3]{L_1}\right)}$$

12. Calculus, 2ADV C3 2017 HSC 16a

i.



Using Pythagoras:

$$L = AE + EB$$

$$= \sqrt{5^2 + x^2} + \sqrt{7^2 + (9-x)^2}$$

$$= \sqrt{25 + x^2} + \sqrt{49 + (9-x)^2} \dots \text{as required}$$

ii. From diagram:

$$\sin \alpha = \frac{x}{\sqrt{25 + x^2}} \text{ and } \sin \beta = \frac{9-x}{\sqrt{49 + (9-x)^2}}$$

♦ Mean mark 41%.

$$L = \sqrt{25 + x^2} + \sqrt{49 + (9-x)^2}$$

$$\frac{dL}{dx} = \frac{2x}{\sqrt{25 + x^2}} - \frac{2(9-x)}{\sqrt{49 + (9-x)^2}}$$

$$\text{If } \frac{dL}{dx} = 0,$$

$$\Rightarrow \frac{2x}{\sqrt{25 + x^2}} = \frac{2(9-x)}{\sqrt{49 + (9-x)^2}}$$

$$\frac{x}{\sqrt{25 + x^2}} = \frac{9-x}{\sqrt{49 + (9-x)^2}}$$

$$\sin \alpha = \sin \beta \dots \text{as required}$$

iii. If $\sin \alpha = \sin \beta$, then $\alpha = \beta$ and

$$\triangle ACE \parallel \triangle BDE$$

Using corresponding sides of similar triangles:

$$\frac{x}{5} = \frac{9-x}{7}$$

$$7x = 45 - 5x$$

$$12x = 45$$

$$\therefore x = \frac{45}{12} \text{ km}$$

♦♦ Mean mark 29%.

- iv. If point B is reflected across the river, AEB will be a straight line.
If any other point is chosen, AEB would not be straight and the distance would be longer.

♦♦♦ Mean mark 9%.

13. Calculus, 2ADV C3 SM-Bank 13

i. In $\triangle BCD$,

$$\cos \theta = \frac{BD}{a}$$

$$\therefore BD = a \cos \theta$$

$$\sin \theta = \frac{CD}{a}$$

$$\therefore CD = a \sin \theta$$

ii. $L = 4 + 2BD + CD + a$

$$= 4 + 2a \cos \theta + a \sin \theta + a$$

$$= 4 + a + 2a \cos \theta + a \sin \theta$$

iii. Noting that a is a constant:

$$\frac{dL}{d\theta} = -2a \sin \theta + a \cos \theta$$

When $\frac{dL}{d\theta} = 0$,

$$-2a \sin \theta + a \cos \theta = 0$$

$$a \cos \theta = 2a \sin \theta$$

$$\therefore BD = 2CD \text{ (using part (a))}$$

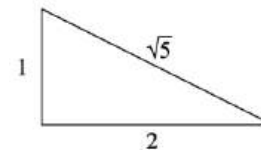
♦ Mean mark (Vic) 35%.

iv. SP's when $\frac{dL}{d\theta} = 0$,

$$-2a \sin \theta + a \cos \theta = 0$$

$$\sin \theta = \frac{1}{2} \cos \theta$$

$$\tan \theta = \frac{1}{2}$$



If $\tan \theta = \frac{1}{2}$, $\cos \theta = \frac{2}{\sqrt{5}}$, $\sin \theta = \frac{1}{\sqrt{5}}$

♦♦♦ Mean mark (Vic) 5%.

$$L_{\max} = 4 + a + 2a \cos \theta + a \sin \theta$$

$$= 4 + (3\sqrt{5}) + 2(3\sqrt{5})\left(\frac{2}{\sqrt{5}}\right) + (3\sqrt{5})\left(\frac{1}{\sqrt{5}}\right)$$

$$= 4 + 3\sqrt{5} + 12 + 3$$

$$= 19 + 3\sqrt{5} \text{ cm}$$

14. Calculus, 2ADV C4 2019 HSC 16c

i. $y = x^r$

$$\frac{dy}{dx} = rx^{r-1}$$

When $x = 1$, $\frac{dy}{dx} = r$

Equation of tangent:

$$y - 1 = r(x - 1)$$

$$y = rx - r + 1$$

When $y = 0$:

$$rx - r + 1 = 0$$

$$rx = r - 1$$

$$x = \frac{r - 1}{r}$$

$\therefore$ Tangent meets x -axis at $\left(\frac{r - 1}{r}, 0\right)$

♦♦ Mean mark part (i) 31%.

ii. Area under curve

$$= \int_0^1 x^r$$

$$= \left[\frac{1}{r + 1} \cdot x^{r+1} \right]_0^1$$

$$= \frac{1}{r + 1} \times 1^{r+1} - 0$$

$$= \frac{1}{r + 1}$$

Area under tangent

$$= \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \left(1 - \frac{r - 1}{r} \right) \times 1$$

$$= \frac{1}{2} \left(1 - \frac{r - 1}{r} \right)$$

♦♦♦ Mean mark part (ii) 21%.

$$\begin{aligned}
 \therefore R &= \frac{1}{r+1} - \frac{1}{2} \left(1 - \frac{r-1}{r} \right) \\
 &= \frac{1}{r+1} - \frac{1}{2r} [r - (r-1)] \\
 &= \frac{1}{r+1} - \frac{1}{2r} \\
 &= \frac{2r - (r+1)}{2r(r+1)} \\
 &= \frac{r-1}{2r(r+1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } R &= \frac{r-1}{2r(r+1)} = \frac{r-1}{2r^2+2r} \\
 \frac{dR}{dr} &= \frac{(2r^2+2r) \times 1 - (r-1)(4r+2)}{(2r^2+2r)^2} \\
 &= \frac{2r^2+2r-4r^2-2r+4r+2}{(2r^2+2r)^2} \\
 &= \frac{-2r^2+4r+2}{(2r^2+2r)^2} \\
 &= \frac{-2(r^2-2r-1)}{(2r^2+2r)^2}
 \end{aligned}$$

Find r when $\frac{dR}{dr} = 0$:

$$r^2 - 2r - 1 = 0$$

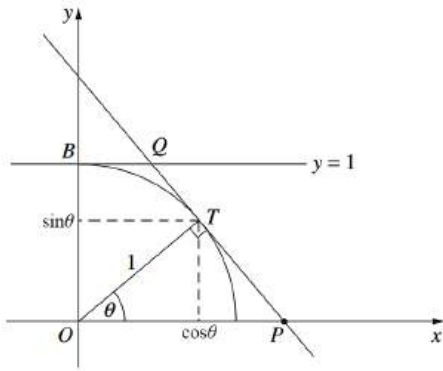
$$\begin{aligned}
 r &= \frac{2 \pm \sqrt{(-2)^2 - 4 \times 1 \times (-1)}}{2} \\
 &= \frac{2 \pm \sqrt{8}}{2} \\
 &= 1 + \sqrt{2} \quad (r \geq 1)
 \end{aligned}$$

r	1	$1 + \sqrt{2}$	3
$\frac{dR}{dr}$	$\frac{1}{4}$	0	$-\frac{1}{144}$

$\therefore R_{\max}$ occurs when $r = 1 + \sqrt{2}$

♦♦ Mean mark part (iii) 23%.

i.

Find T :

$$\text{Since } \cos \theta = \frac{x}{1} \text{ and } \sin \theta = \frac{y}{1}$$

$$\therefore T(\cos \theta, \sin \theta)$$

$$\text{Gradient of } OT = \frac{\sin \theta}{\cos \theta}$$

$$\therefore \text{Gradient } PT = -\frac{\cos \theta}{\sin \theta} \quad (\perp \text{ lines})$$

Equation of PT where

$$m = -\frac{\cos \theta}{\sin \theta}, \text{ and through } (\cos \theta, \sin \theta)$$

$$\text{Using } y - y_1 = m(x - x_1)$$

$$y - \sin \theta = -\frac{\cos \theta}{\sin \theta}(x - \cos \theta)$$

$$y \sin \theta - \sin^2 \theta = -x \cos \theta + \cos^2 \theta$$

$$x \cos \theta + y \sin \theta = \sin^2 \theta + \cos^2 \theta$$

$$x \cos \theta + y \sin \theta = 1 \quad \dots \text{ as required}$$

ii. Find Q :

$$Q \Rightarrow \text{intersection of } x \cos \theta + y \sin \theta = 1 \text{ and } y = 1$$

$$x \cos \theta + \sin \theta = 1$$

$$x \cos \theta = 1 - \sin \theta$$

$$x = \frac{1 - \sin \theta}{\cos \theta}$$

$$\therefore \text{Length of } BQ \text{ is } \frac{1 - \sin \theta}{\cos \theta} \text{ units}$$

$$\text{iii. Show Area } OPQB = \frac{2 - \sin \theta}{2 \cos \theta}$$

$$A = \frac{1}{2}h(a+b) \text{ where } h = OB = 1 \quad a = OP \text{ and}$$

$$b = BQ = \frac{1 - \sin \theta}{\cos \theta}$$

Find length OP :

$$P \Rightarrow x \cos \theta + y \sin \theta = 1 \text{ cuts } x\text{-axis}$$

$$x \cos \theta = 1$$

$$x = \frac{1}{\cos \theta}$$

$$\Rightarrow \text{Length } OP = \frac{1}{\cos \theta}$$

♦♦ Mean mark 24%

$$\begin{aligned} \text{Area } OPQB &= \frac{1}{2} \times 1 \left(\frac{1}{\cos \theta} + \frac{1 - \sin \theta}{\cos \theta} \right) \\ &= \frac{1}{2} \left(\frac{2 - \sin \theta}{\cos \theta} \right) \\ &= \frac{2 - \sin \theta}{2 \cos \theta} \quad \text{u}^2 \quad \dots \text{ as required} \end{aligned}$$

iv. Find θ such that Area $OPQB$ is a MIN

$$A = \frac{2 - \sin \theta}{2 \cos \theta}$$

$$\frac{dA}{d\theta} = \frac{2 \cos \theta (-\cos \theta) - (2 - \sin \theta)(-2 \sin \theta)}{4 \cos^2 \theta}$$

$$= \frac{4 \sin \theta - 2 \sin^2 \theta - 2 \cos^2 \theta}{4 \cos^2 \theta}$$

$$= \frac{4 \sin \theta - 2(\sin^2 \theta + \cos^2 \theta)}{4 \cos^2 \theta}$$

$$= \frac{4 \sin \theta - 2}{4 \cos^2 \theta}$$

$$= \frac{2 \sin \theta - 1}{2 \cos^2 \theta}$$

MAX or MIN when $\frac{dA}{d\theta} = 0$

$$\Rightarrow 2 \sin \theta - 1 = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6} \quad 0 < \theta < \frac{\pi}{2}$$

Test for MAX/MIN

$$\text{If } \theta = \frac{\pi}{12} \quad \frac{dA}{d\theta} < 0$$

$$\text{If } \theta = \frac{\pi}{3} \quad \frac{dA}{d\theta} > 0 \Rightarrow \text{MIN}$$

$\therefore$ Area $OPQB$ is a MIN when $\theta = \frac{\pi}{6}$.

Mean mark 19%

IMPORTANT: Look for any opportunity to use the identity $\sin^2 \theta + \cos^2 \theta = 1 \rightarrow$ often a key to simplifying difficult trig equations.

IMPORTANT: Is the 1st or 2nd derivative test easier here? Examiners have often made one significantly easier than the other.

16. Calculus, 2ADV C3 2013 HSC 14b

i. Need to show $r^2 = 12\,900t^2 - 18\,000t + 10\,000$

$$RB = 100 - 50t$$

$$RA = 80t$$

Using the cosine rule

$$r^2 = (RB)^2 + (RA)^2 - 2(RB)(RA)\cos\angle R$$

$$= (100 - 50t)^2 + (80t)^2 - 2(100 - 50t)(80t)\cos 60$$

$$= 10\,000 - 10\,000t + 2500t^2 + 6400t^2 - 8000t + 4000t^2$$

$$= 12\,900t^2 - 18\,000t + 10\,000 \quad \dots \text{ as required}$$

ii. Max/min when $\frac{dr^2}{dt} = 0$

$$\frac{dr^2}{dt} = 25\,800t - 18\,000 = 0$$

$$t = \frac{18\,000}{25\,800} = \frac{30}{43}$$

$$\text{When } t = \frac{30}{43}, \quad \frac{d^2(r^2)}{dt^2} = 25\,800 > 0 \Rightarrow \text{MIN}$$

$$\therefore \text{ Minimum distance when } t = \frac{30}{43} \text{ hr}$$

$$\text{Find } r \text{ when } t = \frac{30}{43}$$

$$r^2 = 12\,900\left(\frac{30}{43}\right)^2 - 18\,000\left(\frac{30}{43}\right) + 10\,000$$

$$= 3720.9302\dots$$

$$\therefore r = \sqrt{3720.9302\dots}$$

$$\approx 60.99942\dots$$

$$\approx 61 \text{ km (nearest km)}$$

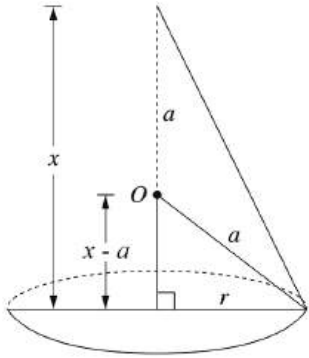
$\therefore$ MIN distance between the cars is 61 km.

♦♦ Mean mark 26%

♦♦ Mean mark 27%

ALGEBRA TIP: Finding the derivative of r^2 (rather than making r the subject), makes calculations ***much easier***. *ENSURE* you apply the test to confirm a minimum.

i. Show $V = \frac{1}{3}\pi(2ax^2 - x^3)$



$$V = \frac{1}{3}\pi r^2 h$$

Using Pythagoras

$$(x - a)^2 + r^2 = a^2$$

$$\begin{aligned} r^2 &= a^2 - (x - a)^2 \\ &= a^2 - x^2 + 2ax - a^2 \\ &= 2ax - x^2 \end{aligned}$$

$$\begin{aligned} \therefore V &= \frac{1}{3} \times \pi \times (2ax - x^2) \times x \\ &= \frac{1}{3}\pi(2ax^2 - x^3) \dots \text{as required} \end{aligned}$$

ii. $\frac{dV}{dx} = \frac{1}{3}\pi(4ax - 3x^2)$

$$\frac{d^2V}{dx^2} = \frac{1}{3}\pi(4a - 6x)$$

Max or min when $\frac{dV}{dx} = 0$

$$\frac{1}{3}\pi(4ax - 3x^2) = 0$$

$$4ax - 3x^2 = 0$$

$$x(4a - 3x) = 0$$

$$3x = 4a, \quad x \neq 0$$

$$x = \frac{4}{3}a$$

When $x = \frac{4}{3}a$

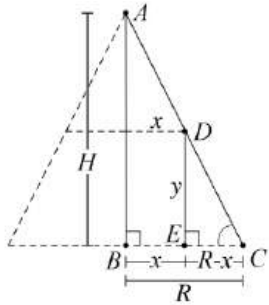
$$\frac{d^2V}{dx^2} = \frac{1}{3}\pi\left(4a - 6 \times \frac{4}{3}a\right)$$

$$= \frac{1}{3}\pi(-4a) < 0$$

$\Rightarrow$ MAX

$\therefore$ Cone volume is a maximum when $x = \frac{4}{3}a$.

i. Show $V = \frac{H}{R}\pi x^2(R - x)$



♦♦ Mean mark 16%.

Consider $\triangle ABC$ and $\triangle DEC$

$$\angle ABC = \angle DEC = 90^\circ$$

$\angle BCA$ is common

$\therefore \triangle ABC \parallel \triangle DEC$ (equiangular)

$$\therefore \frac{DE}{EC} = \frac{AB}{BC} \quad (\text{corresponding sides of similar triangles})$$

$$\frac{y}{R - x} = \frac{H}{R}$$

$$y = \frac{H}{R}(R - x)$$

Volume of cylinder

$$= \pi r^2 h$$

$$= \pi x^2 \times \frac{H}{R}(R - x)$$

$$= \frac{H}{R}\pi x^2(R - x) \dots \text{as required.}$$

ii. $V = \frac{H}{R}\pi x^2(R - x)$

$$\frac{dV}{dx} = \frac{H}{R}\pi[(x^2 \times -1) + 2x(R - x)]$$

$$= \frac{H}{R}\pi[-x^2 + 2xR - 2x^2]$$

$$= \frac{H}{R}\pi[2xR - 3x^2]$$

♦♦ Mean mark 21%.

$$\frac{dV^2}{dx^2} = \frac{H}{R}\pi[2R - 6x]$$

Max or min when $\frac{dV}{dx} = 0$

$$\frac{H}{R}\pi[2xR - 3x^2] = 0$$

$$2xR - 3x^2 = 0$$

$$x(2R - 3x) = 0$$

$$x = 0 \quad \text{or} \quad 3x = 2R$$

$$x = \frac{2R}{3}$$

When $x = 0$

$$\frac{d^2V}{dx^2} = \frac{H}{R}\pi[2R - 0] > 0$$

$\therefore$ Minimum

When $x = \frac{2R}{3}$

$$\frac{d^2V}{dx^2} = \frac{H}{R}\pi\left[2R - 6 \times \frac{2R}{3}\right]$$

$$= \frac{H}{R}\pi[-2R] < 0$$

$\therefore$ Maximum

Maximum Volume of cylinder

$$= \frac{H}{R}\pi\left(\frac{2R}{3}\right)^2\left(R - \frac{2R}{3}\right)$$

$$= \frac{H}{R}\pi \times \frac{4R^2}{9} \times \frac{R}{3}$$

$$= \frac{4\pi HR^2}{27}$$

$$= \frac{\frac{4\pi HR^2}{27}}{\frac{1}{3}\pi R^2 H}$$

$$= \frac{4\pi HR^2}{27} \times \frac{3}{\pi R^2 H}$$

$$= \frac{4}{9}$$

Volume of cone = $\frac{1}{3}\pi R^2 H$

$\therefore \frac{\text{Max Volume of Cylinder}}{\text{Volume of cone}}$

$\therefore$ The volume of the inscribed cylinder does not exceed $\frac{4}{9}$ of the cone volume.